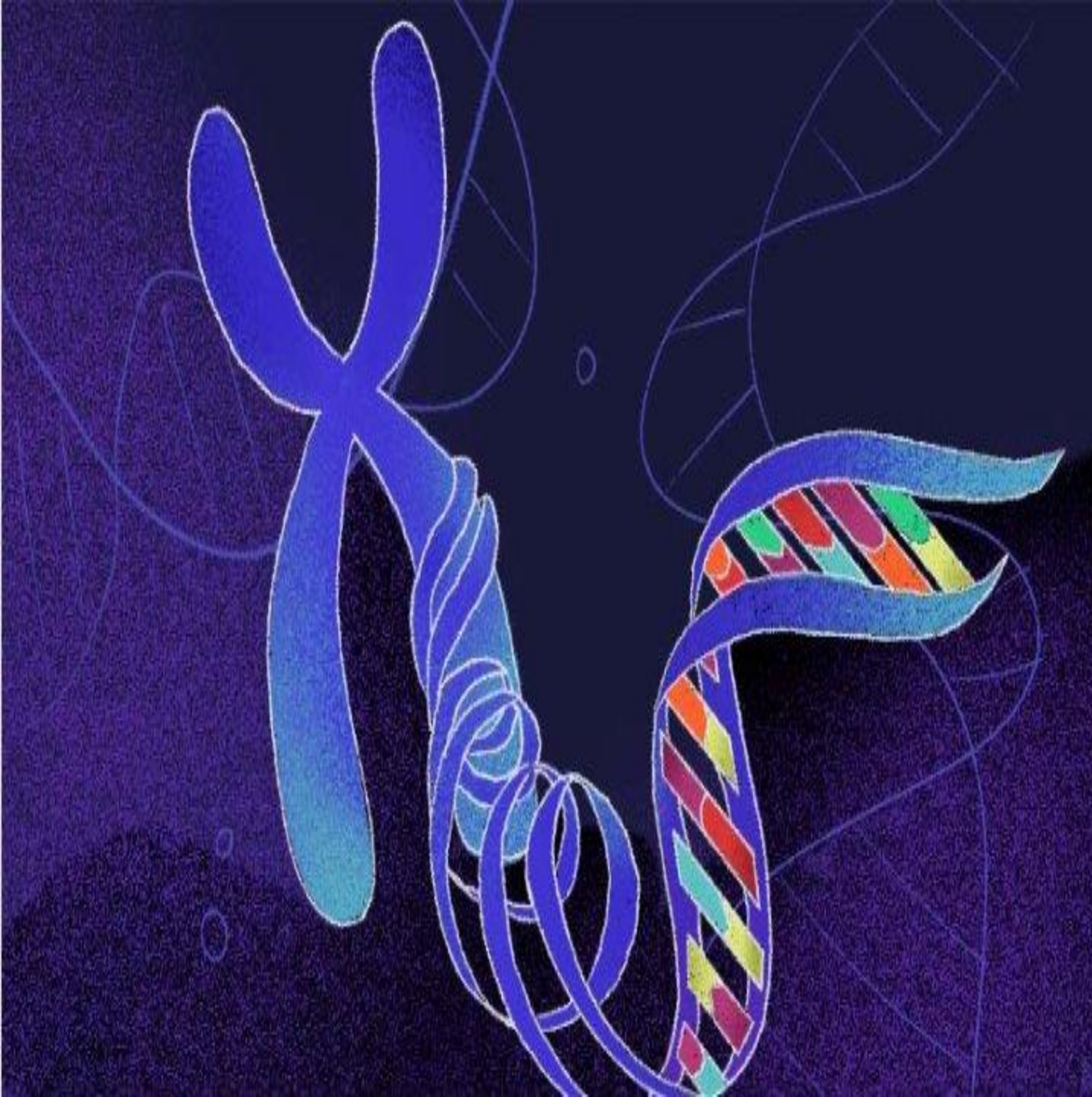




Genetics and Epigenetics

Introduction to Biology

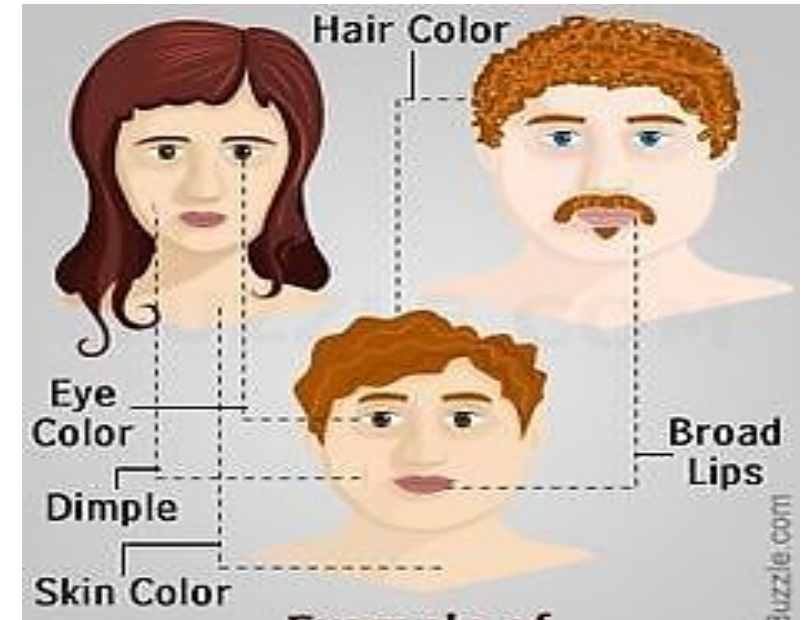
Chapter: 9



Inspiring Excellence

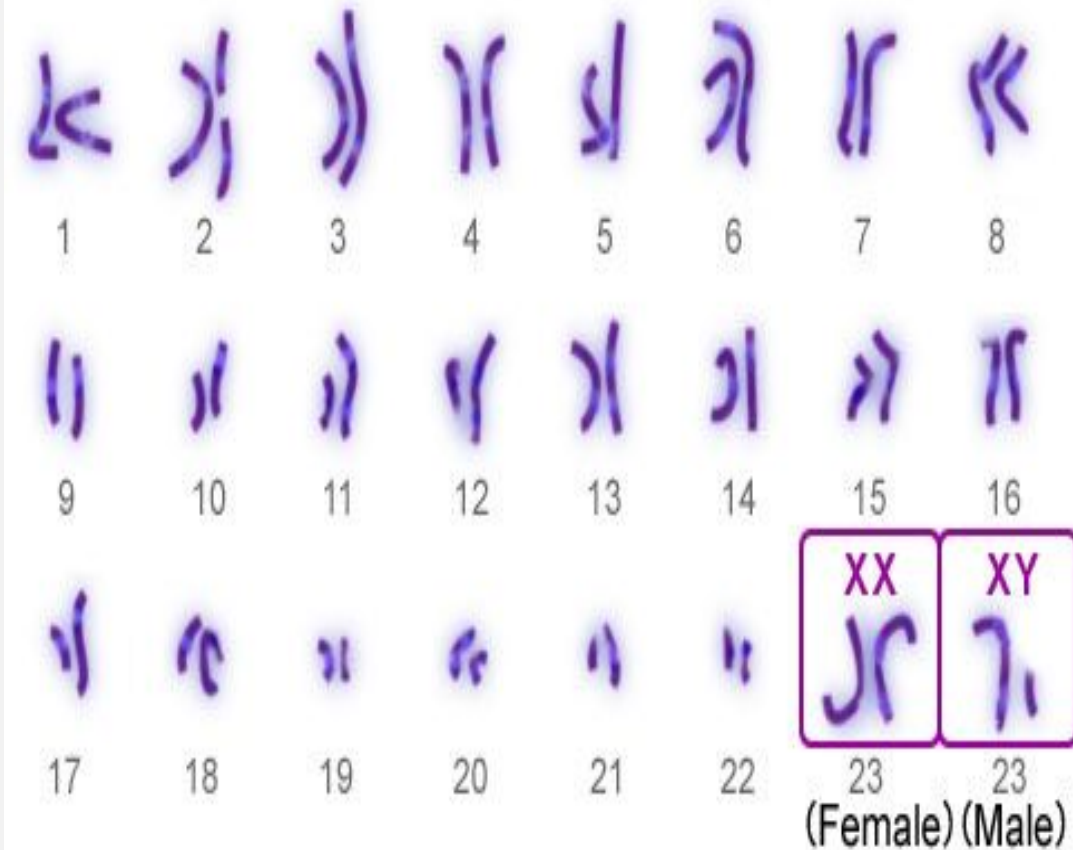
Have you ever wondered why children have similarities to both their parents?

- We acquire these similar characteristics (traits) through a process called **inheritance** (heredity).
- Inheritance occurs through transfer of genetic material from parents to their offspring.



Chromosomes

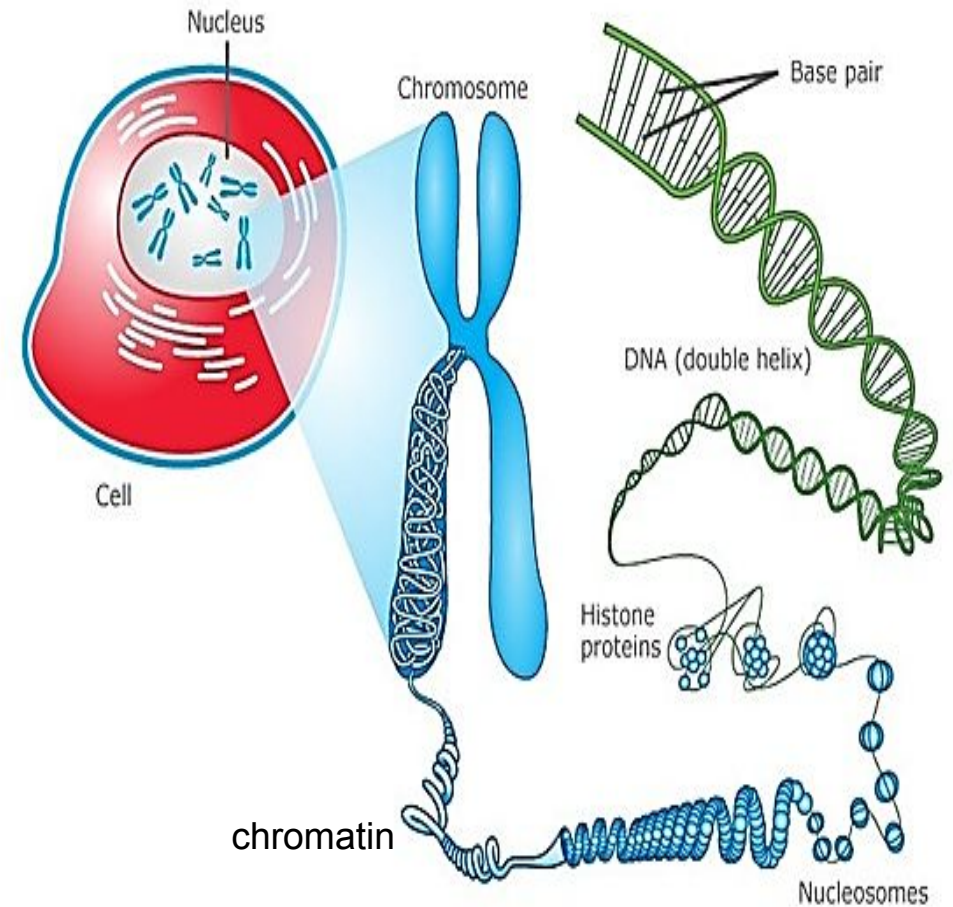
- They are present in the nucleus of the cell (eukaryotes) or present in the cytoplasm (prokaryotes).
- Different organisms have different numbers of chromosomes.
- Humans have 23 pairs of chromosomes (46 in total): 1 from each pair comes from mother and one from father.



Karyotyping

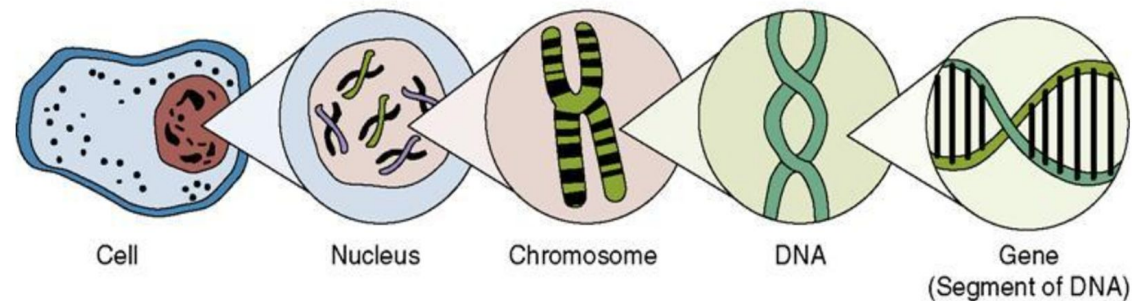
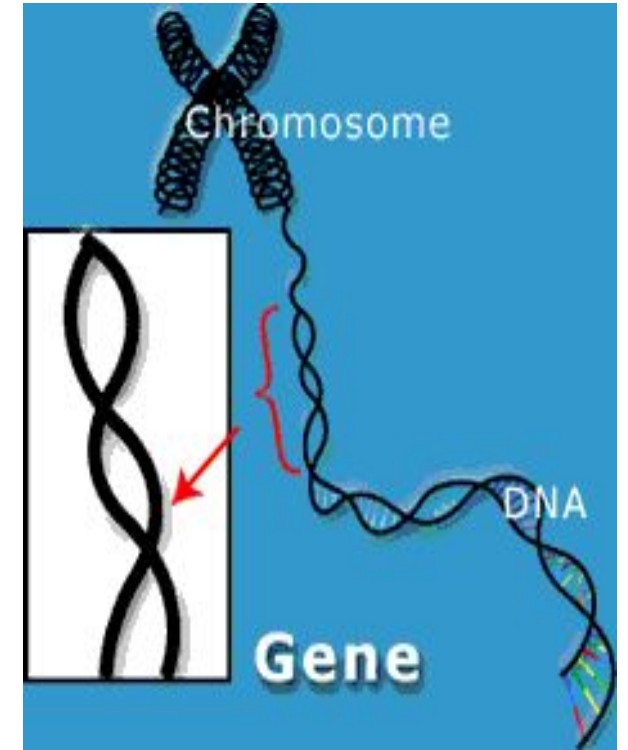
Chromosomes

- DNA are coiled in proteins called **histones** which form structures known as **nucleosomes**.
- Nucleosomes further coil up to form thread like structures, called **chromatins**.
- Chromatins coil and form **chromosomes**.
- A DNA molecule is basically a coiled chromosome!
- **DNA -> (gene) -> nucleosome -> chromatin -> chromosome**



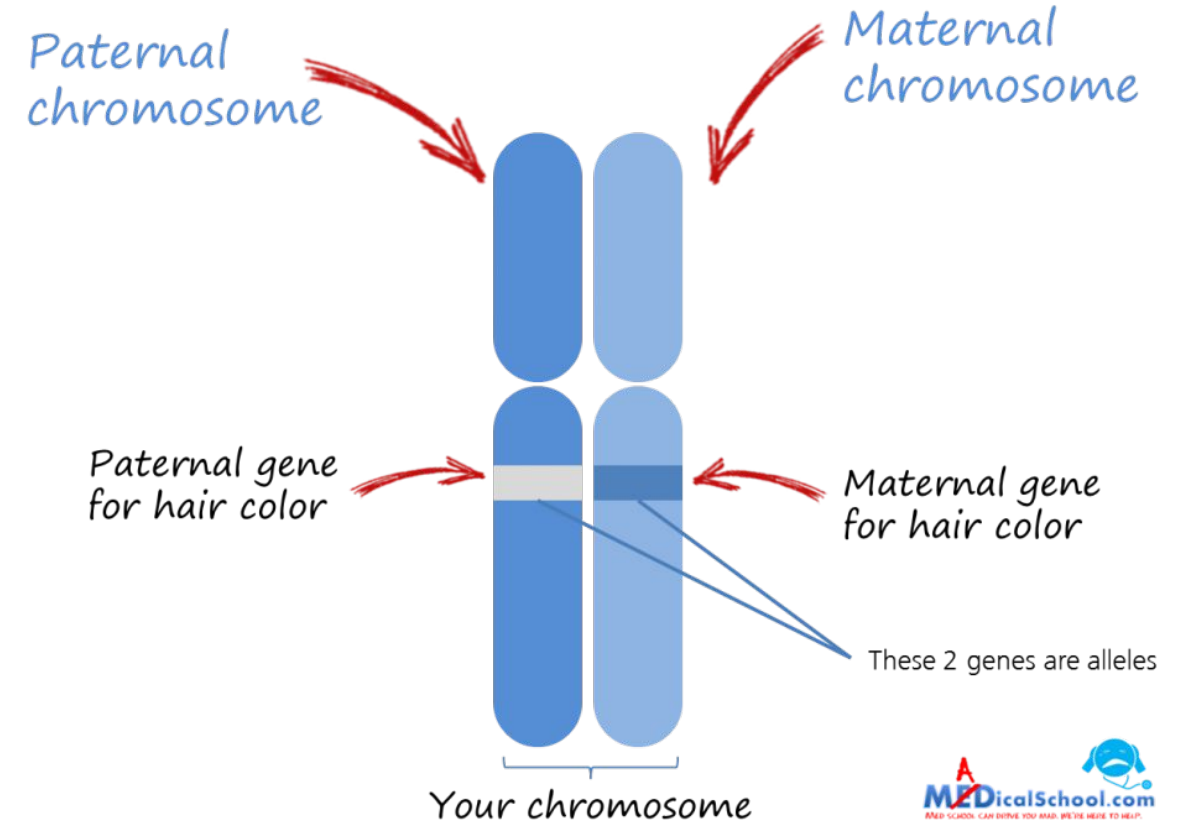
Genes

- Units of information about specific traits
- A sequence of DNA
- Passed from parents to offspring
- The complete set of DNA of an organism is called its **genome**.
- The study of genes, genetic variation and heredity is called **Genetics**.



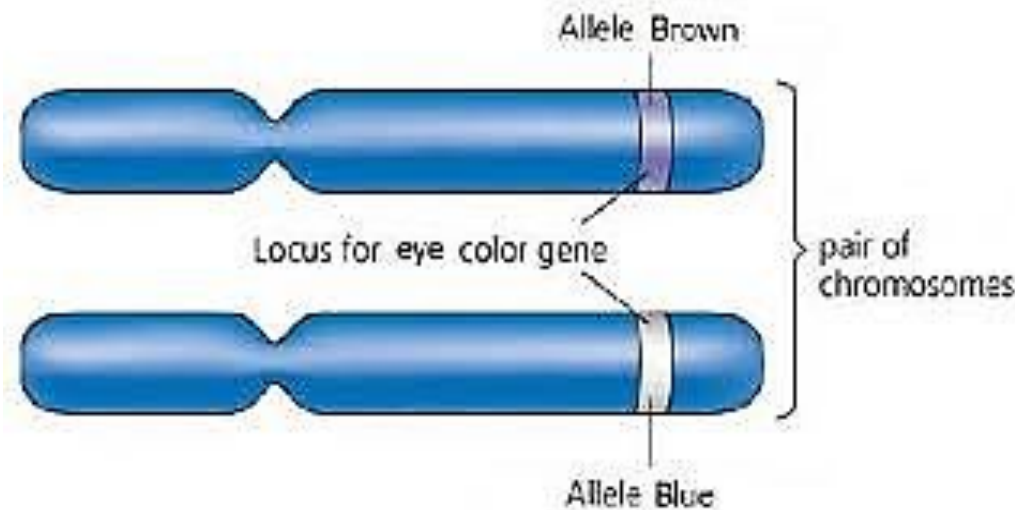
Alleles

- Different molecular forms of the same gene
- Arise by mutation
- Each allele is passed from each parent to the offspring



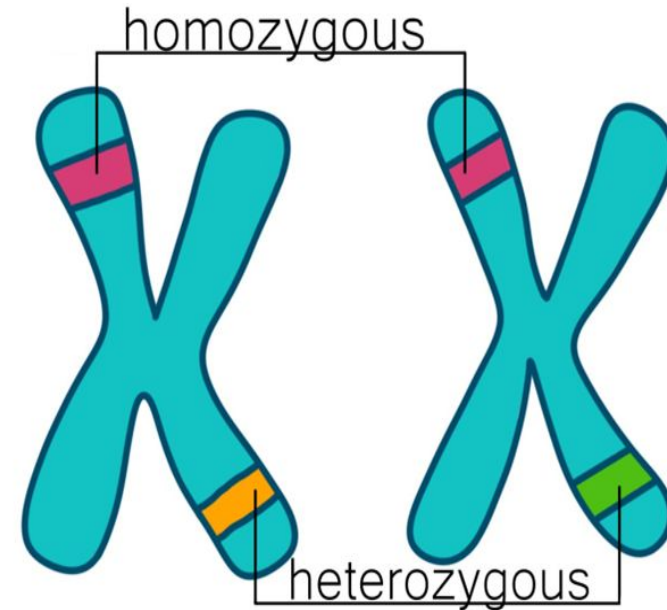
Locus

- Each gene has a specific location (locus) on a chromosome
- Alleles are located in the same locus of two homologous chromosomes



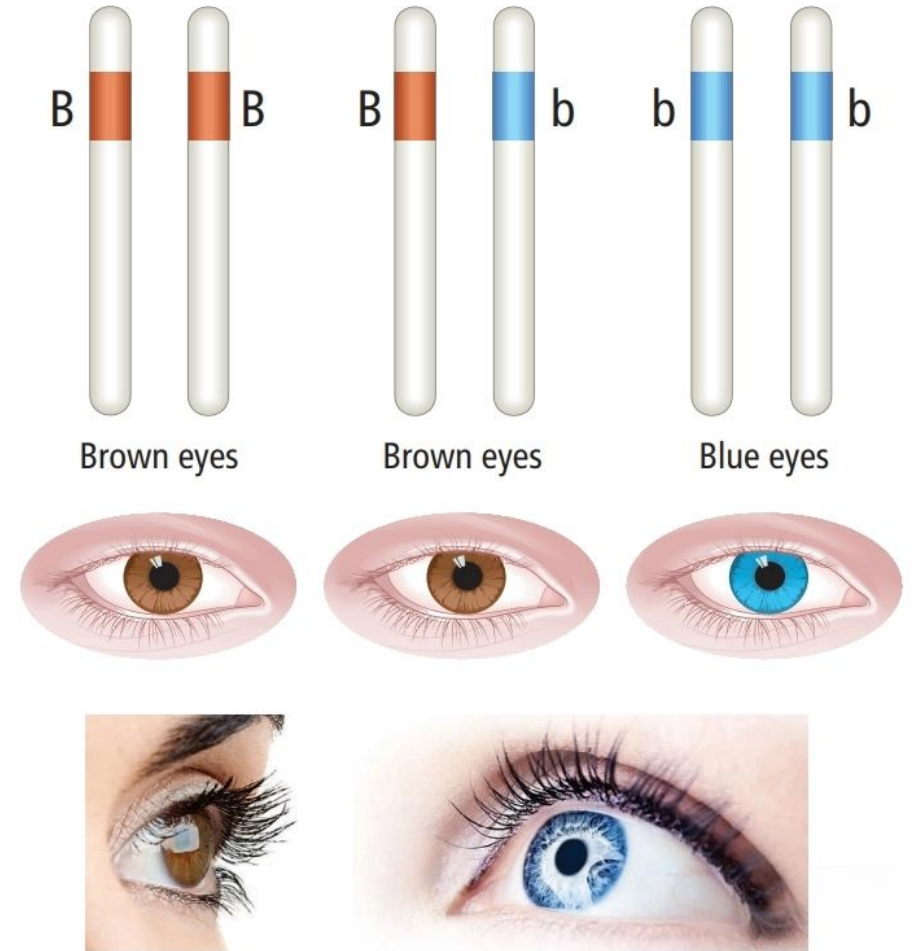
Allele combinations

- Homozygous
 - having two identical alleles at a locus
 - BB or bb
- Heterozygous
 - having two different alleles at a locus
 - Bb



Dominant & Recessive Alleles

- A dominant allele is always expressed and hence masks a recessive allele that is paired with it.
- Recessive alleles are only expressed when both alleles are recessive.
- Dominant alleles are written in capital letters (B) and recessive alleles are written in small letters (b).



Allele

1 dominant + 1
dominant

1 recessive + 1
recessive

1 dominant + 1
recessive

Term

Homozygous dominant
(BB)

Homozygous recessive
(bb)

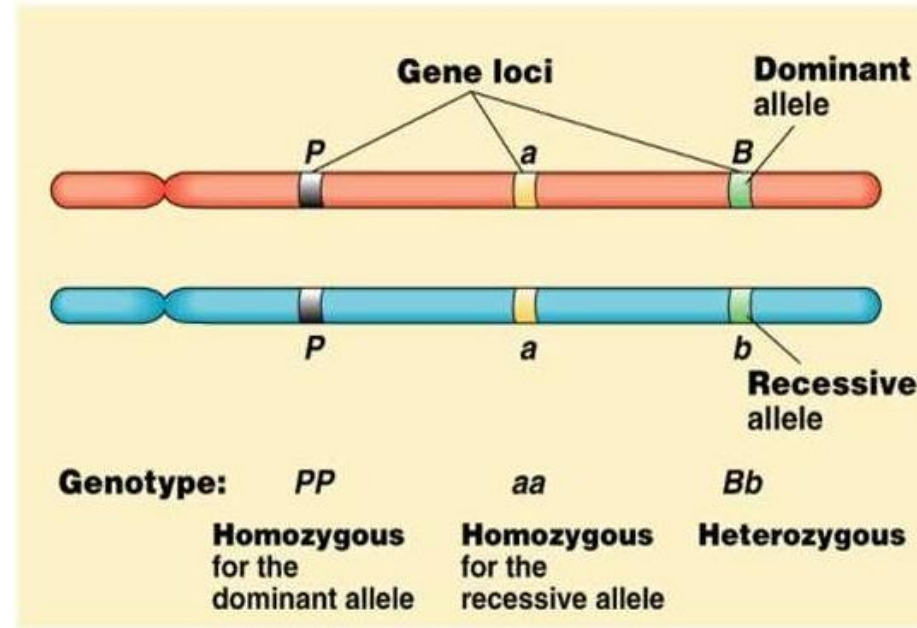
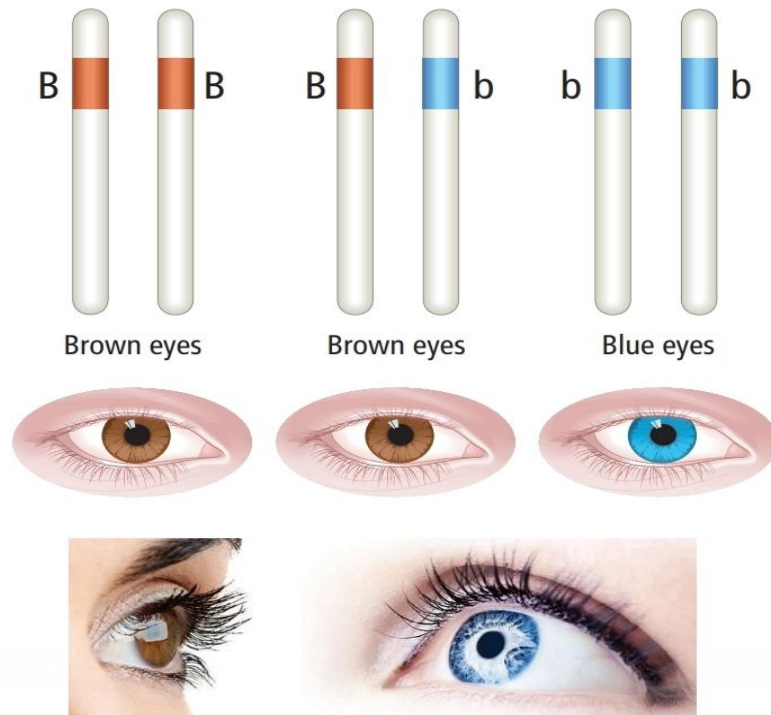
Heterozygous dominant
(Bb)

Result

Dominant trait is
expressed

Recessive trait is
expressed

Dominant trait is
expressed

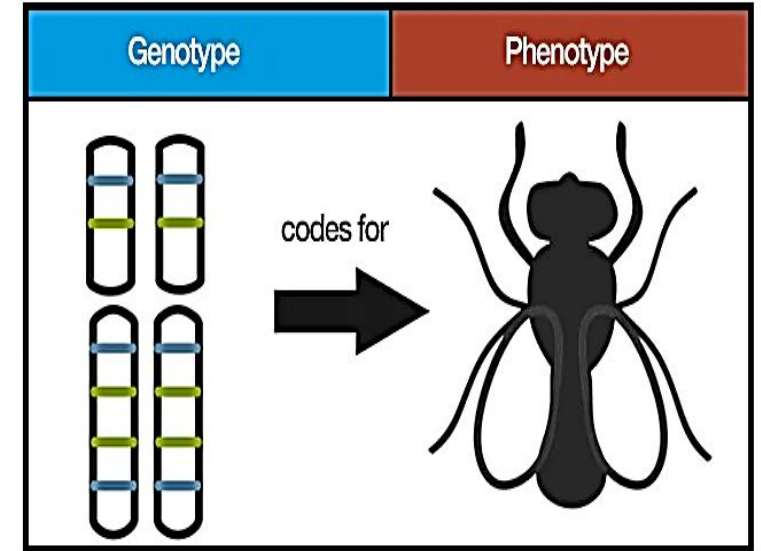


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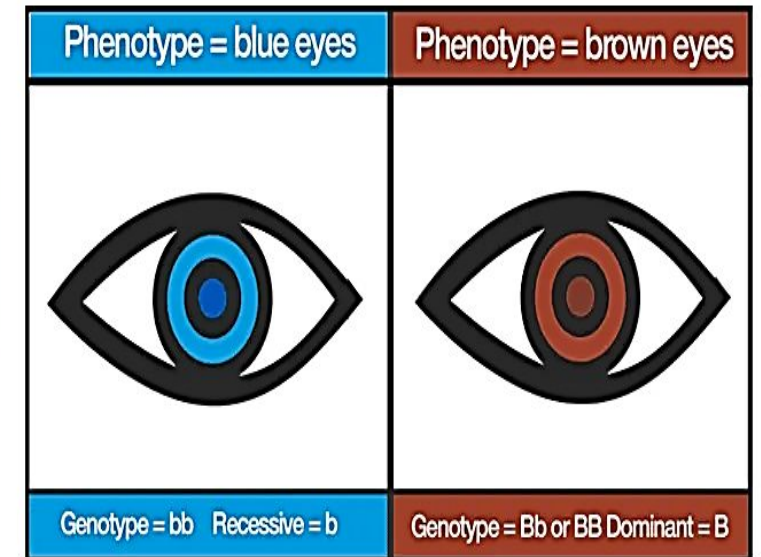
Genotype & Phenotype

- Genotype refers to particular genes an individual carries
- Phenotype refers to an individual's observable traits
- We cannot always determine genotype by observing phenotype: Homozygous dominant or Heterozygous?

Genotype is the genetic makeup of the organism



Phenotype is the physical appearance of the organism



GENOTYPE VERSUS PHENOTYPE

Genotype is the genetic makeup of an organism

Can be determined by observing DNA by genotyping methods

Completely depends on the gene sequences

Inherited by the offspring

Consists of all hereditary information that is the expressed and suppressed genes

Phenotype is the morphology, properties and behavior of an organism

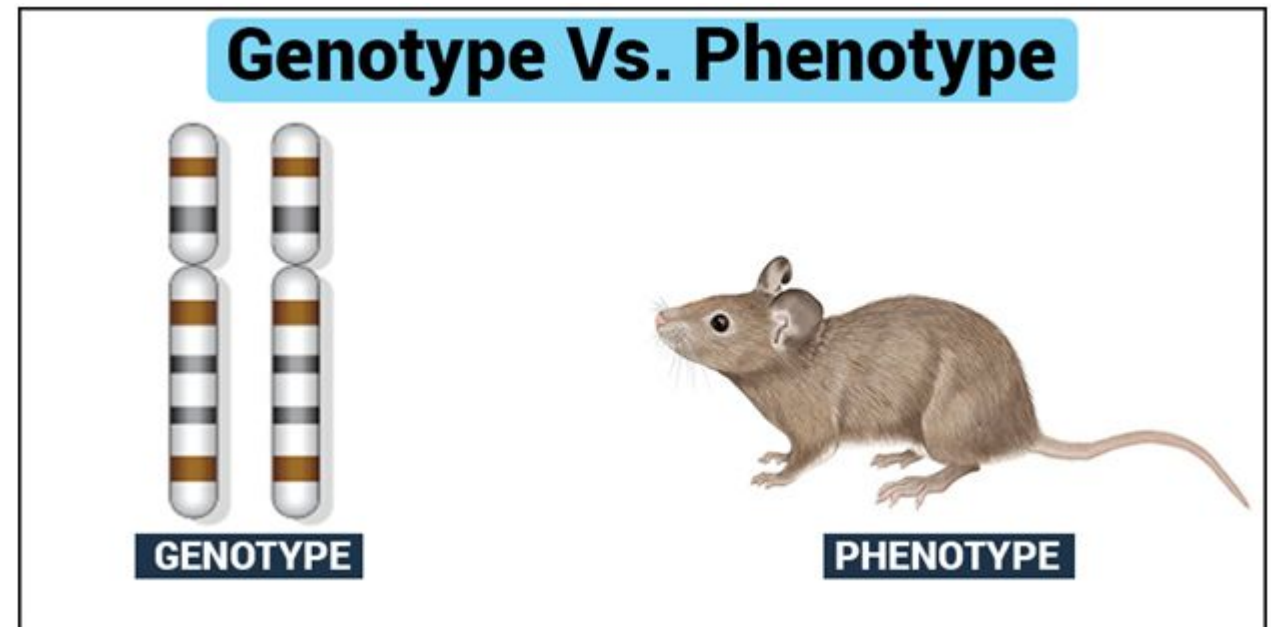
Can be determined by observing outward characters

Depends on the genotype and environmental factors

Not inherited by the offspring

Consists of expressed genes

Visit www.pediaa.com



Gregor Mendel

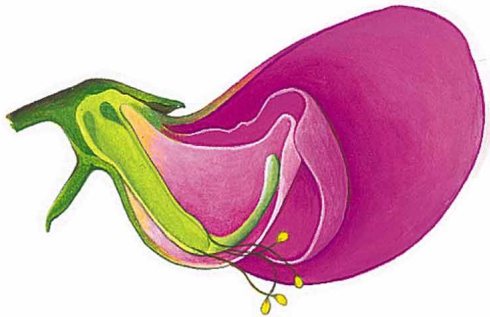
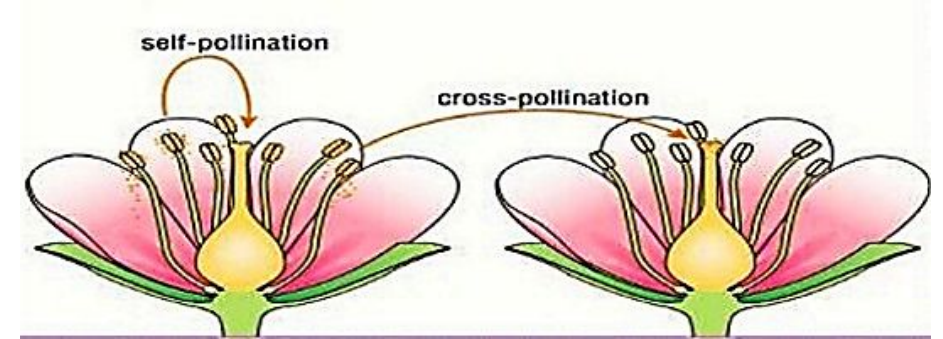
- Strong background in plant breeding and mathematics.
- Using pea plants, found indirect but observable evidence of how parents transmit genes to offspring.
- Particulate theory: physical traits are inherited as “particles”.



Father of genetics!

The Garden Pea Plant

- Self-pollinating
- True breeding (different alleles not normally introduced): usually homozygous for traits
- Can be experimentally cross-pollinated
- Distinct contrasting characters



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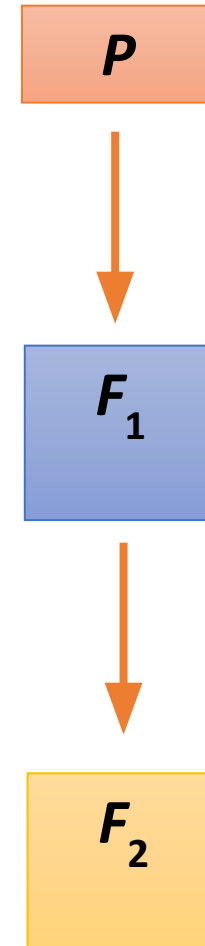
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Tracking Generations

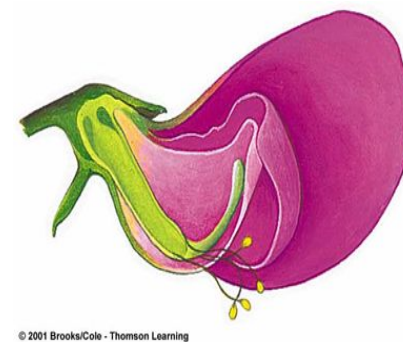
- Parental generation mates to produce:
- First-generation offspring mate to produce:
- Second-generation offspring



Monohybrid cross: focuses on one trait!

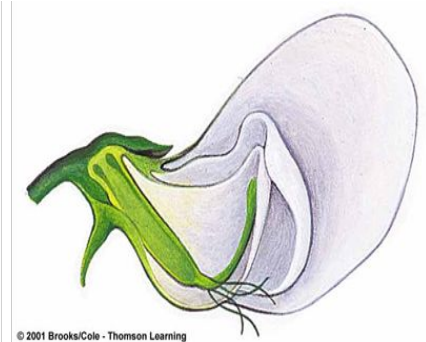
- Mono: one
- Hybrid: combination of two organisms
- For garden pea plants, Mendel observed the color of the flower petals -> trait
- A mix of genetics between two **homozygous genotypes**:
- Either completely dominant or completely recessive

		Female gametes	
		<i>a</i>	<i>a</i>
Male gametes	<i>A</i>	<i>Aa</i>	<i>Aa</i>
	<i>a</i>	<i>Aa</i>	<i>Aa</i>



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Purple: AA



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White: aa

F_1 Results of Monohybrid Cross

Result

All offspring were **heterozygous dominant (Aa)**

True-breeding
homozygous recessive
parent plant



aa

True-breeding
homozygous dominant
parent plant



AA

A

A

Aa	Aa
Aa	Aa

a

a

F_1
PHENOTYPES



Aa



Aa



Aa

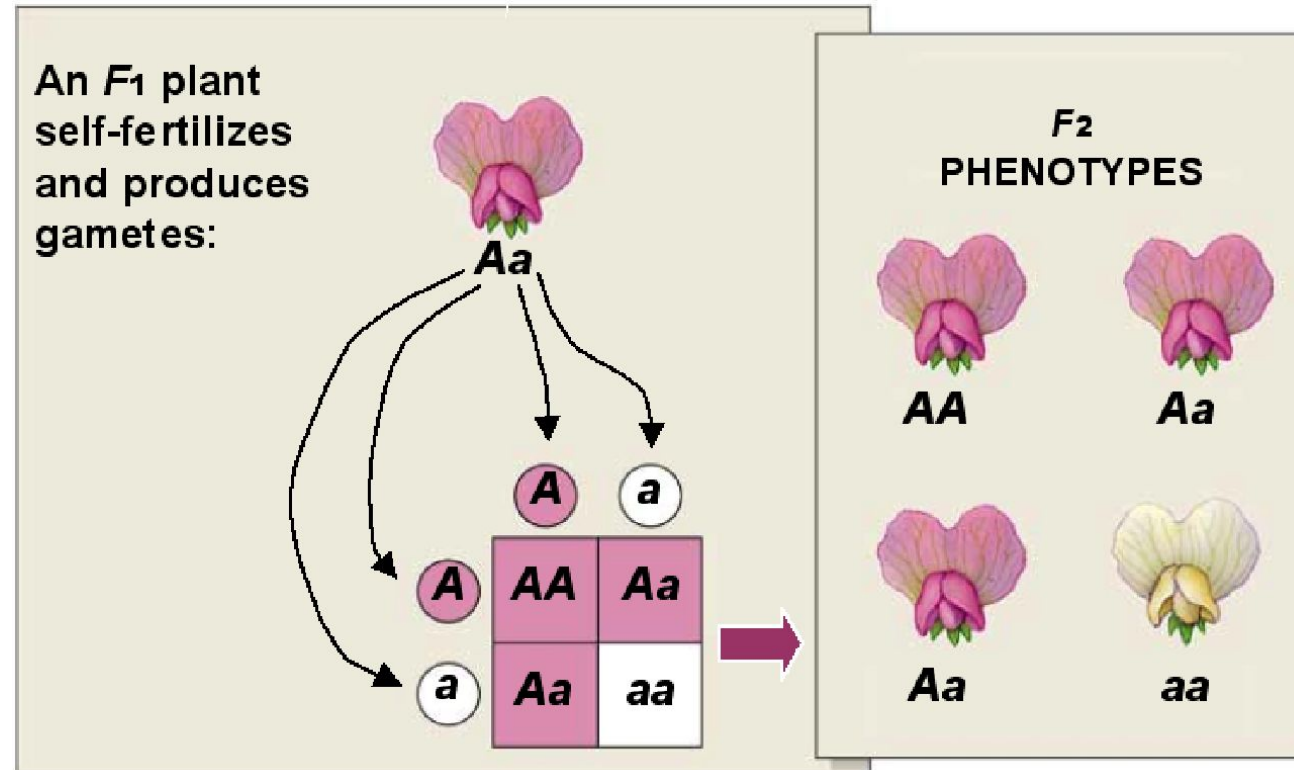


Aa

F_2 Results of Monohybrid Cross

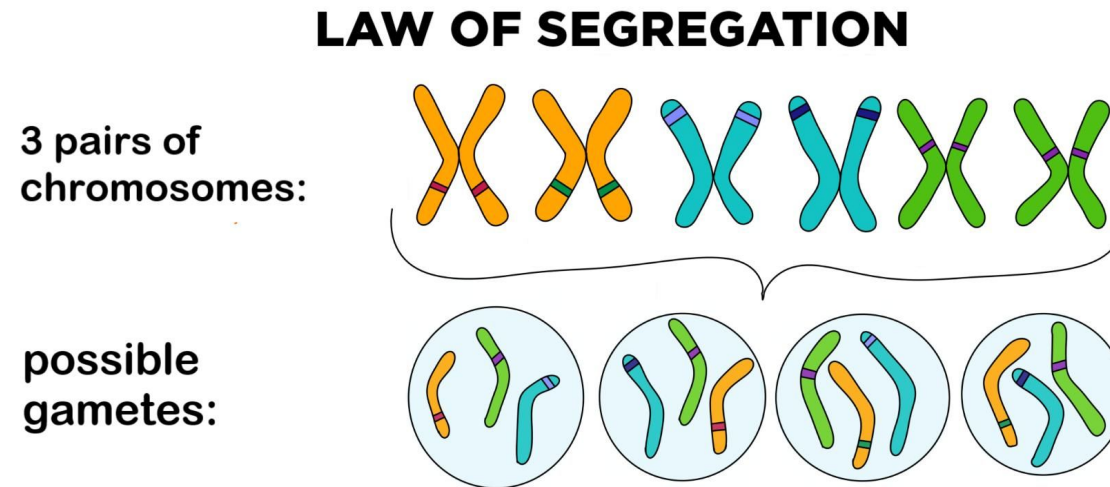
Result

- 2 offspring were **heterozygous dominant (Aa)**
- 1 offspring was **homozygous dominant (AA)**
- 1 offspring was **homozygous recessive (aa)**
- Phenotypic ratio – 3 dominant : 1 recessive



Mendel's Theory of Segregation

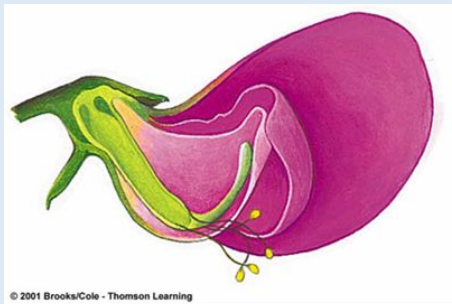
- An individual inherits a unit of information (allele) about a trait from each parent.
- During gamete formation, the alleles segregate from each other.



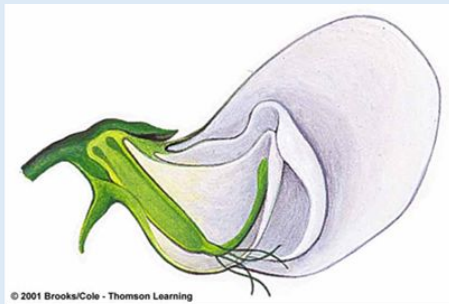
The 2 copies of each gene separate and end up in different gametes

Dihybrid Cross: focuses on two traits!

- Di = two
- Hybrid = combination of two organisms
- Experimental cross between individuals that are homozygous for **different versions** of **two** traits.
- Alleles of the two traits, located on different homologous chromosome pairs.



Purple: AA



White: aa

+



Tall: BB



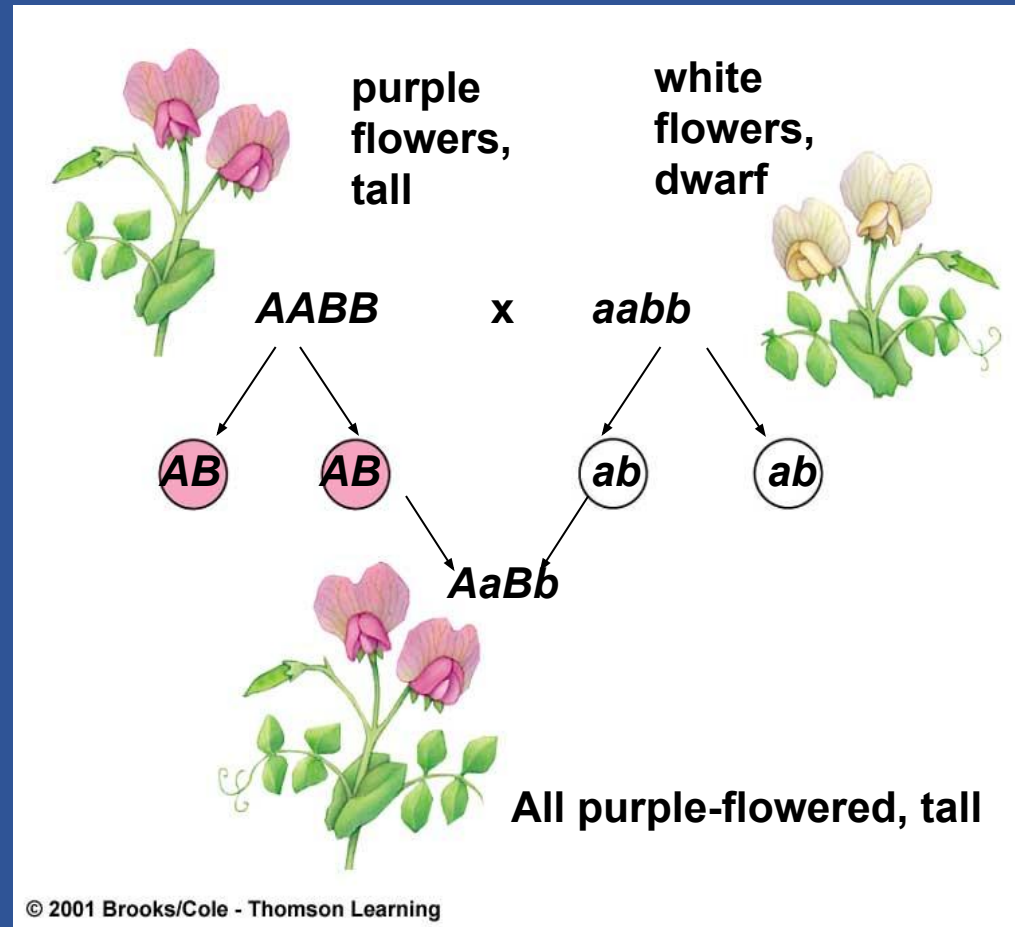
Dwarf: bb

A Dihybrid Cross - F₁ Results

TRUE-BREEDING PARENTS:

GAMETES:

F₁ HYBRID OFFSPRING:



F_1 Results of Mendel's Dihybrid Crosses

$AABB \times aabb$

	ab	ab	ab	ab
AB	$AaBb$	$AaBb$	$AaBb$	$AaBb$
AB	$AaBb$	$AaBb$	$AaBb$	$AaBb$
AB	$AaBb$	$AaBb$	$AaBb$	$AaBb$
AB	$AaBb$	$AaBb$	$AaBb$	$AaBb$

- All plants displays **the dominant form of both traits** (Purple and tall)
- We now know:
 - All plants inherited one allele for each trait from each parent
 - All plants were heterozygous ($AaBb$)

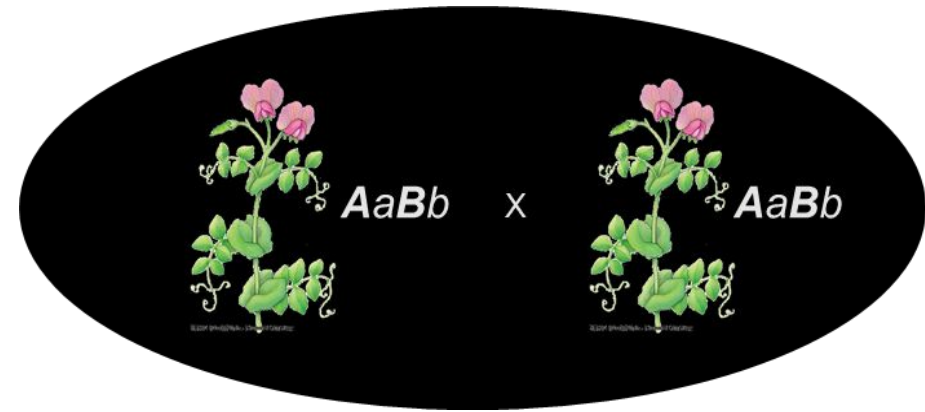
Phenotypic Ratios in F_2

If the alleles of the two traits are on **separate homologous chromosome pairs**, sixteen allele combinations are possible:

Four Phenotypes:

- Purple-flowered, tall (9/16)
- White-flowered, tall (3/16)
- Purple-flowered, dwarf (3/16)
- White-flowered, dwarf (1/16)

Phenotypic ratio: 9:3:3:1



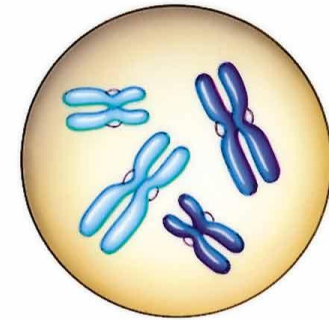
	$\frac{1}{4}$ AB	$\frac{1}{4}$ Ab	$\frac{1}{4}$ aB	$\frac{1}{4}$ ab
$\frac{1}{4}$ AB	$\frac{1}{16}$ $AABB$	$\frac{1}{16}$ $AABb$	$\frac{1}{16}$ $AaBB$	$\frac{1}{16}$ $AaBb$
$\frac{1}{4}$ Ab	$\frac{1}{16}$ $AABb$	$\frac{1}{16}$ $AAbb$	$\frac{1}{16}$ $AaBb$	$\frac{1}{16}$ $Aabb$
$\frac{1}{4}$ aB	$\frac{1}{16}$ $AaBB$	$\frac{1}{16}$ $AaBb$	$\frac{1}{16}$ $aaBB$	$\frac{1}{16}$ $aaBb$
$\frac{1}{4}$ ab	$\frac{1}{16}$ $AaBb$	$\frac{1}{16}$ $Aabb$	$\frac{1}{16}$ $aaBb$	$\frac{1}{16}$ $aabb$

Mendel's Law of Independent Assortment

- Mendel concluded that the two “units” (alleles) for the first trait were to be assorted into gametes independently of the two “units” (alleles) for the other trait.
- Members of each pair of homologous chromosomes are sorted into gametes at random during meiosis.

Independent Assortment

AaBb

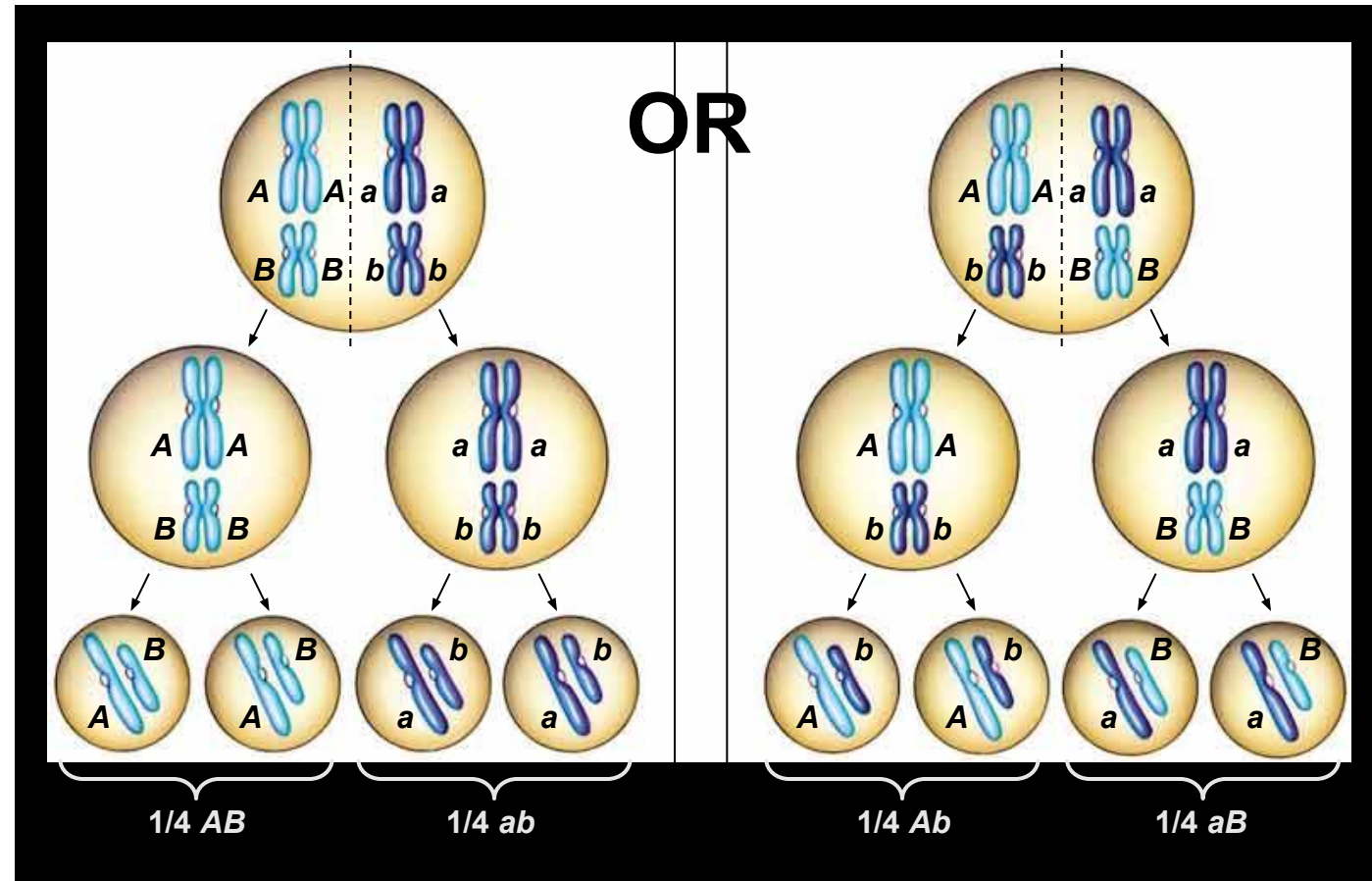


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Metaphase I

Metaphase II:

Gametes:



MENDEL'S FIRST LAW
VERSUS
MENDEL'S SECOND LAW

MENDEL'S FIRST LAW

A principle that describes the separation of the two copies of each hereditary factor during the production of gametes

Also called the law of segregation

Uses a monohybrid cross

Ratio of offspring is 3:1

MENDEL'S SECOND LAW

A principle that describes the independent assortment of alleles of different genes during the formation of gametes

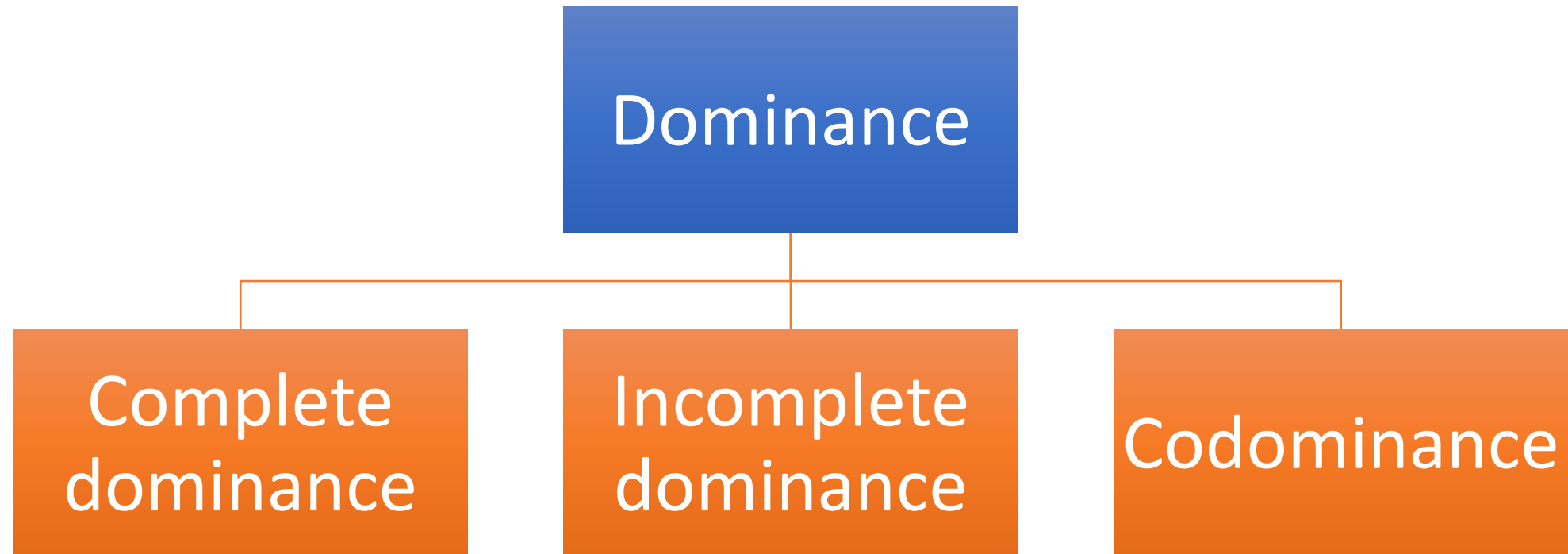
Also called the law of independent assortment

Uses a dihybrid cross

Ratio of offspring is 9:3:3:1

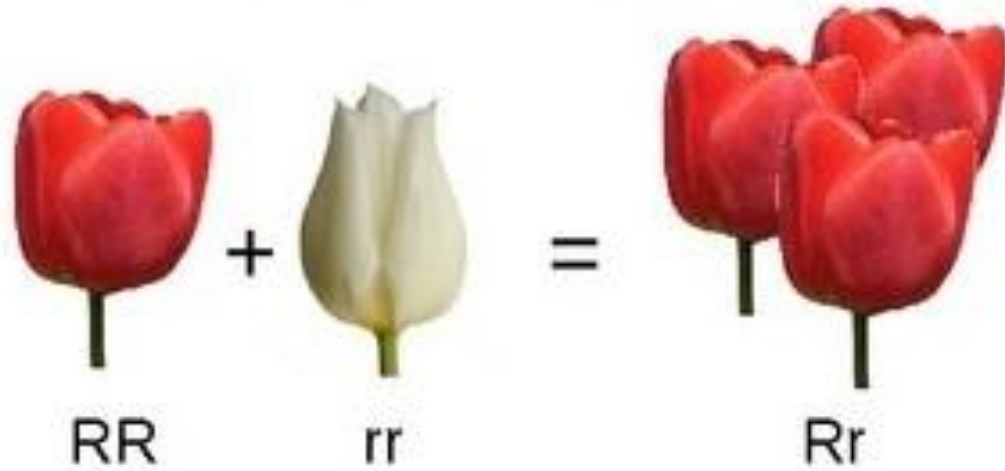
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Dominance Relations



Complete dominance

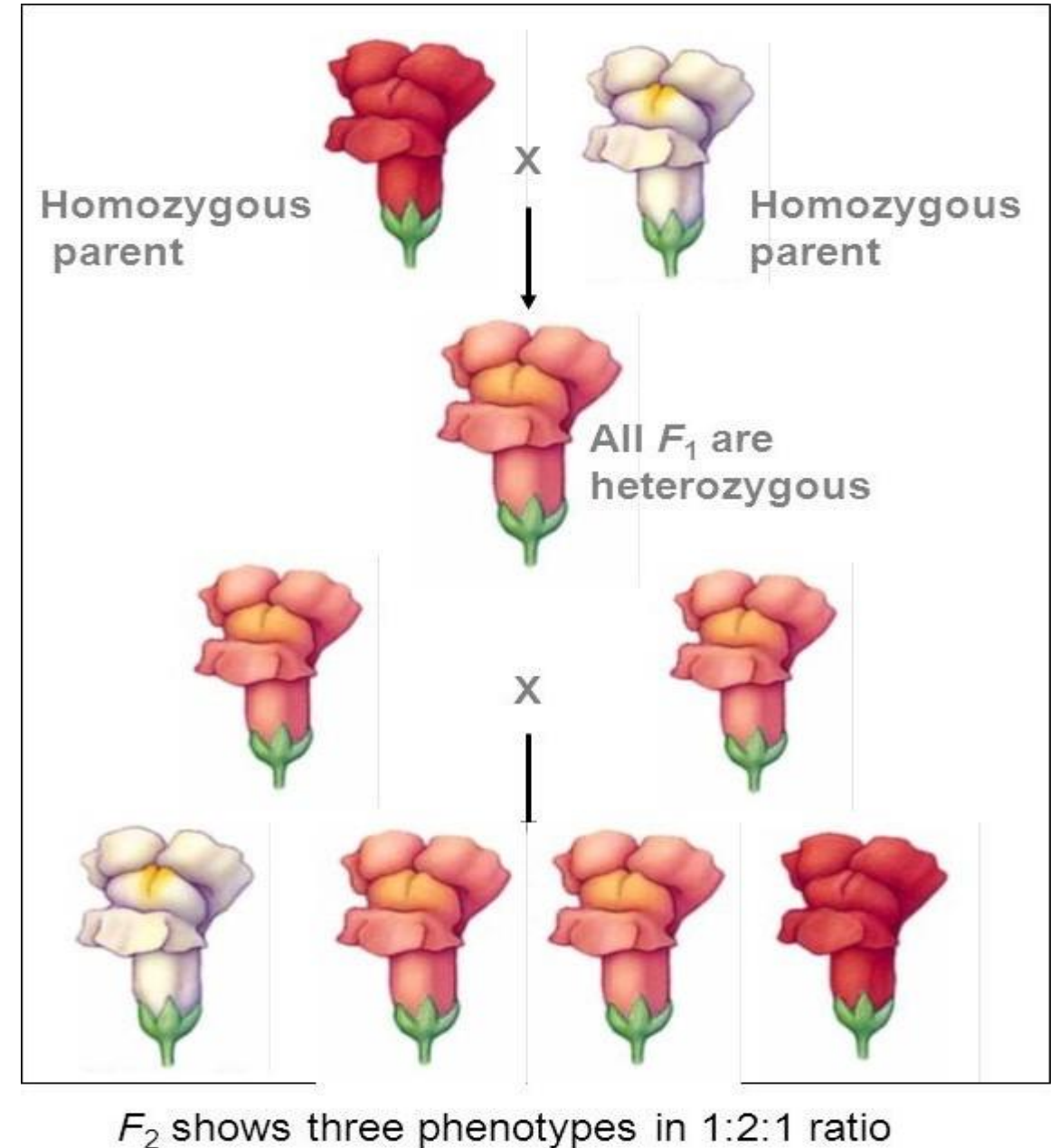
- Dominance in heterozygous condition where the dominant allele completely masks the effects of the recessive allele.



Incomplete dominance

- Dominant allele does not completely mask the effect of the recessive allele.
- Heterozygous phenotype is somewhere between that of two homozygotes.
- Blend of both traits

Flower colour in Snapdragons



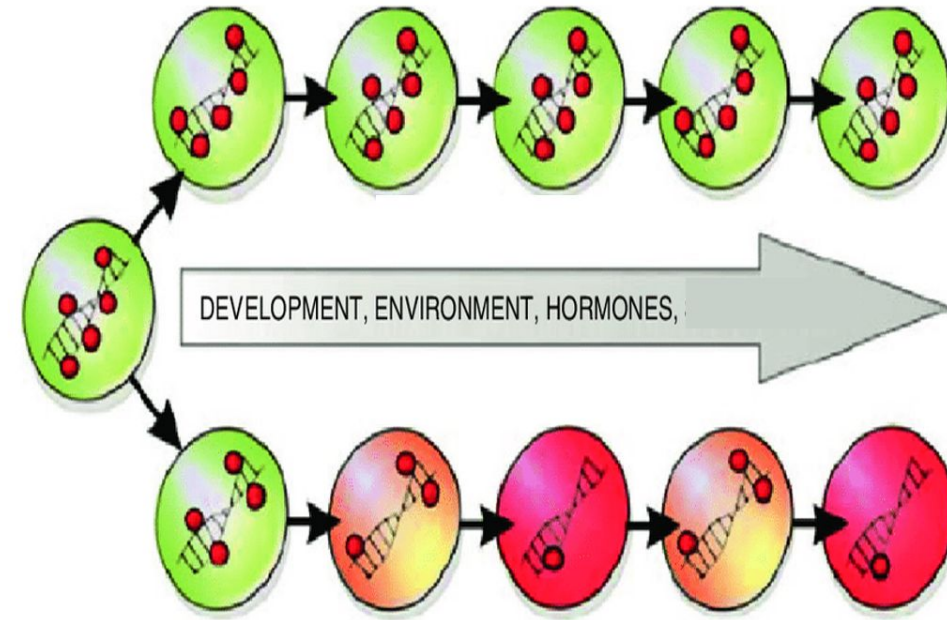
Codominance

- Both alleles of a particular trait are fully expressed.
- Non-identical alleles specify two phenotypes that are both expressed in heterozygous offsprings.
- Offspring has a phenotype that shows combination of both traits.



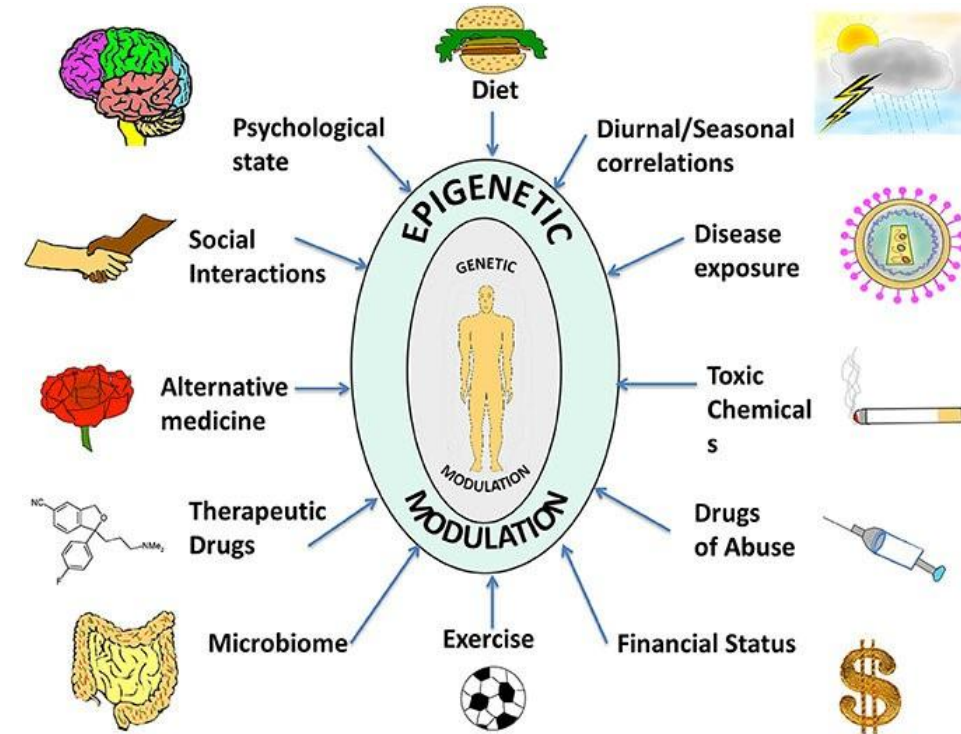
Epigenetics

- Epigenetics is the study of how environmental influence can affect the expression of genes.
- Epigenetic changes can determine how much a gene is expressed. Can also switch certain genes “on” or “off”!
- During development, DNA accumulates chemical marks known as **epigenomes**. Epigenomes determine how much/ how little of the genes is expressed.
 - For e.g., genetically identical twins (**same genotype**) can exhibit different behaviors, skills, health etc. (**different phenotype**) because of the different experiences they have.
 - **NOT Mutation**, DNA sequence is not changed! Only how it is read!



Epigenetics Factors

- **Genetics:** The arrangement of genes and protein structures themselves can sometimes influence how the chemical marks are distributed across the DNA.
- **Environment:** Outside factors that can have an impact on genes. For e.g. pollutants, heavy metals, electromagnetic radiations etc.
- **Development:** Development can include all stages of life, starting from embryonic, to early childhood, all the way to adulthood.
 - Diet, Psychological state, Social behavior pattern, Disease exposure etc.



Thank you!

